Viva Question Bank

Indian Railway Trains ML Project

Classification and Clustering of Indian Railway Trains Using Machine Learning

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Contents:

Section A - Project Deep Dive (kya, kaise, kaha, kaunsa cell)
Section B - 40 Viva Questions with difficulty tags, examples, aur keywords

Difficulty key: [MOST IMP] = zaroor, [IMPORTANT] = hona chahiye, [MEDIUM] = achha, [EASY] = basic, [HARD] = agar deep pooche.

SECTION A - PROJECT DEEP DIVE

Viva se pehle poora project ek baar sar se paanv tak samajh le. Ye section sabse pehle padh - iske baad Q&A apne aap fit baithega. Hinglish mein likha hai, examples project se hi diye hain.

A1. Project ek line mein

Indian Railways ke trains ka data le kar Machine Learning se do kaam kiye: (1) classify - naye train ko 3 classes (Pass / Exp / SF) mein daalna, (2) cluster - bina label ke trains ko similar groups mein baantna.

A2. Project kya solve karta hai?

- Railway board ko naye train ka type decide karna hota hai. Hum predict kar sakte hain ki naya train Pass hoga, Exp hoga ya SF.
- Operations team similar trains ko group kar sakti hai - schedule optimize karne ke liye.
- Passengers ko rough idea milta hai ki kaunse route pe kaisi service hai.

A3. Data ki kahani

- Source: Kaggle - sripaadsrinivasan/indian-railways-dataset
- Original: trains.json (GeoJSON, 14 MB)
- Converted: trains.csv (636 KB) - notebook CSV-first padhta hai
- Raw records: 5,208 -> Filtered: 4,466 (sirf 3 classes)
- One-hot ke baad 26 features
- Train/Test: 80/20 stratified -> 3,572 train / 894 test

A4. Folder structure - kaha kya rakha hai

```
Indian-Railway-Train/
|-- data/
|   |-- trains.json                <- original raw
|   |-- trains.csv                 <- converted (notebook isko padhta hai)
|   `-- trains_with_clusters.csv   <- clustering output (labels + features)
|
|-- notebooks/
|   `-- main_notebook.ipynb        <- MAIN file, 4 phases isi mein
|
|-- extras/
|   |-- advanced_ml.ipynb          <- CV, feature importance, error analysis
|   `-- README.md
|
|-- models/
|   |-- scaler.joblib              <- fitted StandardScaler
|   `-- dt_classifier.joblib       <- trained Decision Tree
|
|-- visualizations/                <- 20+ PNG plots
|
|-- reports/
|   |-- images/                    <- AI-generated hero images
|   |-- generate_ppt.py
```

```
|   |-- generate_viva_bank.py       <- ye script
|   |-- Viva_Question_Bank.pdf      <- OUTPUT
|   `-- final_report.md
|
|-- .gitignore
|-- MASTER_OPERATING_PROMPT.md
|-- requirements.txt
`-- README.md
```

A5. Notebook - kaunsa cell kya karta hai

Phase 1 - EDA (Exploratory Data Analysis)

- Cell 1-3: Imports + CSV-first load. Agar CSV nahi mila toh JSON fallback.
- Cell 4-6: head, shape, dtypes, nulls, describe - basic sanity check.
- Cell 7-9: class count plot, zone distribution bar, histograms.
- Cell 10-12: bivariate scatter, boxplots, correlation heatmap, pairplot.
- Constant-column audit: har row mein same value wale columns drop.

Phase 2 - Preprocessing + Evaluation

- Duration fix cell: total_duration = duration_h * 60 + duration_m (killer fix).
- One-hot encoding: train_type, zones, etc. 0/1 columns mein.
- name aur number drop - identifier hain, ML mein kaam ke nahi.
- train_test_split(stratify=y) - pehle split, phir scale (leakage nahi).
- Metrics: accuracy, precision, recall, F1 (macro + weighted + micro), CM, ROC-AUC.

Phase 3 - Supervised models

- Cell 30-FIX: Decision Tree - acc 0.9150, macro F1 0.8896 <- WINNER
- KNN (k=5), SVM (RBF), Naive Bayes - comparison ke liye.
- DummyClassifier baseline - 0.5503 accuracy (sirf 'Pass' bolke).
- Model comparison leaderboard + feature importance plot.

Phase 4 - Clustering

- Cell 31-FIX: K-Means k=5 - inertia 69,115.75, silhouette 0.2024.
- Elbow curve se k=5 chuna.
- PCA 2D - 26 features ko 2D mein project kiya.
- DBSCAN (eps=7, min_samples=20) - 4 clusters, silhouette 0.2543, 0.4% noise.
- trains_with_clusters.csv - 3,572 rows, 28 columns.

Extras notebook (advanced_ml.ipynb)

- Mini rebuild cell - poora pipeline chhote version mein.
- 5-fold CV F1 macro 0.8743 +/- 0.0155.
- Feature importance - third_ac top pe (0.355).
- Error analysis chart.
- Models persisted - scaler.joblib, dt_classifier.joblib.

A6. Visualizations folder - har PNG kya dikhati hai

- eda_histograms.png - distance, duration ka distribution.

- eda_class_zone_distribution.png - kaunsa zone mein kitne trains, kaunsi class.
- bivariate_scatter.png - distance vs duration scatter.
- boxplots.png - har class ke feature ka spread (outliers).
- correlation_heatmap.png - features ke beech correlation.
- pairplot.png - har pair ka scatter, class colours.
- line_plot_trends.png - duration vs distance trend.
- density_plots.png - KDE curves.
- histograms_logscale.png - skew features ka log-scale view.
- before_after_duration_fix.png - killer slide, fix se pehle vs baad.
- confusion_matrix_best.png - DT ka confusion matrix.
- roc_curves.png - 3 classes ke ROC + AUC.
- model_comparison.png - 5 models ka bar chart.
- feature_importance.png - DT ki top features.
- cross_validation.png - 5 folds ka F1.
- error_analysis.png - kaha model galat hua.
- elbow_curve.png - k vs inertia, k=5 pe kink.
- clusters_pca.png - K-Means ka 2D PCA.
- clusters_k5_fixed_pca.png - duration fix ke baad K-Means PCA.
- dbscan_vs_kmeans.png - dono ka side-by-side.

A7. Models folder - kya saved hai

- scaler.joblib - fitted StandardScaler. Naye data pe same scaling apply karne ke liye.
- dt_classifier.joblib - trained Decision Tree. Load karo, predict karo.

A8. Poora pipeline ek line mein

Raw JSON -> CSV -> 3 classes filter -> EDA -> duration fix -> one-hot -> 80/20 split -> scale -> 5 models -> DT jeeta (91.5%) -> K-Means + DBSCAN -> clusters CSV save -> models persist.

SECTION B - VIVA Q&A

Har question ke saath difficulty tag hai: [MOST IMP] zaroor padhna, [IMPORTANT] achha hai, [MEDIUM]/[EASY] baad mein, [HARD] agar guide deep pooch le. Answer Hinglish mein hai, saath mein project ka example aur keywords diye hain. Keywords match hone chahiye - wording nahi.

Q1. Project ka title kya hai aur ye kya karta hai?

[MOST IMP]

Ans: Title hai 'Classification and Clustering of Indian Railway Trains Using Machine Learning'. Isme Indian Railways ke trains ko do tarike se study karte hain: (1) Supervised Classification - naye train ko 3 classes mein daalna (Pass, Exp, SF), (2) Unsupervised Clustering - bina label ke similar trains ko groups mein baantna.

Example (hamare project se): Jaise school mein 3 sections A, B, C hote hain - waise hi humara model naye train ko dekh kar bata deta hai ki wo Pass hai, Exp hai ya SF.

Keywords: Classification, Clustering, Supervised, Unsupervised, Pass/Exp/SF

Q2. Ye project kyu chuna? Real-world relevance kya hai?

[IMPORTANT]

Ans: Indian Railways duniya ka 4th largest network hai, data bahut rich hai. Railway board naye train ka type decide karta hai, operations team schedule optimize karti hai. Ye real, non-toy dataset hai - textbook wala iris flower nahi.

Example (hamare project se): Agar naya train Mumbai-Delhi chalana hai, uski speed, distance, stops dekh kar predict kar sakte hain ki Pass hoga, Exp hoga ya SF.

Keywords: Real-world, non-toy dataset, operations, scheduling

Q3. Dataset kaha se liya? Kitne rows? Kaunse columns?

[MOST IMP]

Ans: Kaggle se - 'sripaadsrinivasan/indian-railways-dataset'. Original file trains.json (GeoJSON, 14 MB) thi. CSV mein convert ki - 5,208 raw records, 3 classes pe filter kiya - 4,466 trains. One-hot ke baad 26 features. Important columns: train_type, distance, duration_h, duration_m, zones, third_ac, chair_car, first_ac, sleeper.

Example (hamare project se): Humne har zone ke trains cover kiye - ER, NFR, WR, SR etc.

Keywords: Kaggle, GeoJSON, 5,208 raw, 4,466 filtered, 26 features

Q4. Teen classes kya hain? Distribution kya thi?

[IMPORTANT]

Ans: Pass (Passenger), Exp (Express), SF (Superfast). Distribution: Pass 2,459 (55.06%), Exp 1,288 (28.84%), SF 719 (16.10%). Moderate imbalance hai - Pass bahut zyada, SF kam.

Example (hamare project se): Kyunki Pass sabse zyada tha, humne macro F1 ko primary metric banaya, accuracy ko nahi - warna model sirf Pass predict karke 55% le aata.

Keywords: Class distribution, imbalance, 55/29/16

Q5. Class imbalance tha? Handle kaise kiya?

[IMPORTANT]

Ans: Haan, moderate (55/29/16). Humne: (1) accuracy ki jagah macro F1 primary metric banaya -

minority class (SF) bhi respect kare, (2) stratified split - train aur test dono mein same ratio, (3) confusion matrix se per-class error dekha.

Example (hamare project se): Baseline DummyClassifier ne 0.5503 accuracy di - sirf 'Pass' bolke. Lekin macro F1 sirf 0.2367. Isse pata chala accuracy dhoka de sakti hai.

Keywords: Macro F1, stratified split, confusion matrix, minority class

Q6. Data leakage kya hai? Aapne kaise roka?

[MOST IMP]

Ans: Data leakage tab hota hai jab test data ki info training mein chali jaye. Model ratta maar leta hai aur real world mein fail. Humne roka: pehle train_test_split, phir StandardScaler ko sirf train pe FIT kiya, test pe sirf TRANSFORM. Agar pehle scale karte phir split, to test ke mean/std training mein leak ho jaate.

Example (hamare project se): Rule yaad rakho: 'fit' sirf train pe, 'transform' train aur test dono pe.

Keywords: Data leakage, fit, transform, train_test_split order

Q7. name aur number columns drop kyu kiye?

[IMPORTANT]

Ans: Ye identifier hain - har train ka naam aur number unique hai. Model ke liye useless kyunki pattern nahi sikhate, sirf yaad karte hain. Zyada columns se overfitting bhi badhta hai.

Example (hamare project se): Jaise roll number se kisi student ki personality nahi pata chalti - waise hi train number se type predict nahi hota.

Keywords: Identifier, overfitting, feature selection

Q8. duration_m ka kya problem tha? Fix kaise kiya?

[MOST IMP]

Ans: duration_m column sirf minute-component tha (0-58 range), total minutes nahi. Matlab agar train 5 ghante 30 min chalti hai, to duration_m mein 30 aata, 330 nahi. Isse model confuse ho raha tha. Fix: total_duration = duration_h * 60 + duration_m.

Example (hamare project se): Fix se pehle DT ka F1 0.8309 tha, fix ke baad 0.8896 - 0.0587 ka jump!

Keywords: duration_m, total_duration, feature engineering, data quality

Q9. Fix ka impact kya padha? (before vs after)

[MOST IMP]

Ans: Sab 5 supervised models ka F1 badha, aur silhouette bhi. Ye project ka sabse bada finding hai: Data quality > Algorithm choice. KNN +0.0334, DT +0.0587, DT(depth=10) +0.0739, Naive Bayes +0.0129, SVM +0.0483, Silhouette +0.0269.

Example (hamare project se): Koi bhi algorithm badalne se itna consistent jump nahi aata. Data fix karna hi asli kaam tha.

Keywords: Data quality > Algorithm choice, consistent improvement

Q10. Scaling split ke baad kyu kiya, pehle nahi?

[IMPORTANT]

Ans: Kyunki leakage rokna tha. Agar pehle scale karte, test set ke mean/std bhi training data ke saath mix ho jaate. Isliye pehle train_test_split, phir StandardScaler.fit(X_train), phir scaler.transform(X_train aur X_test).

Example (hamare project se): Agar test data pehle se dekha, to evaluation fake ho jayega - real world mein aisa nahi hota.

Keywords: Leakage prevention, scaling order

Q11. StandardScaler kaam kya karta hai?

[EASY]

Ans: Har feature ko is formula se transform karta hai: z = (x - mean) / std. Result: mean 0, std 1. Isse sab features same scale pe aa jaate hain.

Example (hamare project se): distance hazaar mein hai, duration_h 10 mein - bina scale, distance dominate kar deta KNN/SVM mein. Scale ke baad dono barabar.

Keywords: StandardScaler, z-score, mean 0 std 1

Q12. One-hot encoding kya hai? Kyu use kiya?

[MEDIUM]

Ans: Categorical column ko 0/1 columns mein todna. Example: zone column mein 'ER', 'WR' hai - one-hot se 'zone_ER' aur 'zone_WR' do columns ban jate hain, jisme 0/1 aata hai. ML model sirf numbers samajhta hai, strings nahi.

Example (hamare project se): Humne train_type, zones etc. ko one-hot kiya. Isse 26 features ban gaye.

Keywords: One-hot encoding, categorical to numeric, dummy variables

Q13. KNN aur SVM ke liye scaling kyu chahiye, DT ke liye nahi?

[IMPORTANT]

Ans: KNN distance calculate karta hai - bade scale wala feature dominate kar deta. SVM margin bhi distance pe based hai. Isliye scaling zaroori. Decision Tree splits pe kaam karta hai - 'x > 500?' - threshold se farak nahi padta ki value 500 hai ya 500000. Isliye DT scale-free hai.

Example (hamare project se): Humne DT ko unscaled aur scaled dono pe chalaya - same result. Lekin KNN/SVM ka result scale ke bina bahut kharab tha.

Keywords: Distance-based, tree-based, scale-invariant, threshold

Q14. Naive Bayes ka independence assumption kya hai?

[MEDIUM]

Ans: NB maan leta hai ki saare features ek doosre se independent hain - matlab 'distance' aur 'duration' ka koi relation nahi. Ye assumption real mein galat hoti hai (duration distance pe depend karta hai), isliye NB weak perform karta hai.

Example (hamare project se): Hamare project mein NB ka F1 sirf 0.3098 nikla - sabse kam. Kyunki features correlated the.

Keywords: Conditional independence, Bayes theorem, correlated features

Q15. Confusion matrix kya dikhata hai? Apna result batao.

[IMPORTANT]

Ans: 4 cheezein: True Positive, True Negative, False Positive, False Negative. Multi-class mein har class ke liye. Diagonal = correct predictions.

Example (hamare project se): Hamare DT ka overall error 8.50%. Dominant confusion: SF <-> Exp (25 SF ko Exp bola, 16 Exp ko SF bola). Pass sirf 3% galat, Exp 14%, SF 17.4%. Kyunki SF aur Exp features overlap karte hain.

Keywords: TP, TN, FP, FN, diagonal, per-class error

Q16. Accuracy vs F1 - kab kaunsa use karein?

[MOST IMP]

Ans: Accuracy tab jab classes balanced ho. F1 tab jab imbalance ho, kyunki F1 precision aur recall ka harmonic mean hai - minority class ko bhi respect karta hai. Hamare project mein imbalance tha, isliye macro F1 primary metric tha.

Example (hamare project se): Hamara baseline 0.55 accuracy le raha tha sirf 'Pass' bolke, par macro F1 0.2367 thi. Accuracy dhoka de rahi thi, F1 ne sach bataya.

Keywords: Accuracy, precision, recall, F1, harmonic mean, macro

Q17. Best model kaunsa nikla? Kyu jeeta?

[MOST IMP]

Ans: Decision Tree (default parameters) jeeta. Accuracy 91.50%, macro F1 0.8896, weighted F1 0.9146. Kyu: (1) DT ko scaling ki zaroorat nahi, (2) non-linear splits handle karta hai, (3) rule-based - interpretable, (4) one-hot features pe natural fit hai.

Example (hamare project se): KNN 0.8523, SVM 0.8412 - dono DT se kam. Naive Bayes 0.3702 - independence assumption ne maara.

Keywords: Decision Tree, 91.50%, 0.8896 macro F1, interpretable, non-linear

Q18. Top feature importance kaunse nikle?

[IMPORTANT]

Ans: Decision Tree ki importance: (1) third_ac - 0.355, (2) distance - 0.223, (3) total_duration - 0.180, (4) chair_car - 0.118, (5) first_ac - 0.013.

Example (hamare project se): third_ac top pe hai kyunki SF aur Exp trains mein 3AC coaches zyada hote hain, Pass mein kam. Matlab coach composition hi class ka signal hai.

Keywords: Feature importance, third_ac, distance, total_duration

Q19. ROC-AUC kya measure karta hai?

[IMPORTANT]

Ans: ROC curve - True Positive Rate vs False Positive Rate, har threshold pe. AUC = area under curve. 0.5 = random, 1.0 = perfect. Ye batata hai ki model threshold se independent kitna achha separate karta hai.

Example (hamare project se): Hamare results: KNN 0.9355 (best), SVM 0.9285, DT 0.9204. Sab 0.9 ke upar - matlab classes kaafi separable hain.

Keywords: ROC, AUC, TPR, FPR, threshold-independent

Q20. Stratified split kya hai? Kyu use kiya?

[MEDIUM]

Ans: train_test_split mein 'stratify=y' - train aur test dono mein class ratio same rahega. Hamare 55/29/16 ka ratio dono sets mein preserve hua.

Example (hamare project se): Agar random split karte, to test set mein SF 5% hi reh jaata - model ka SF performance judge nahi kar pate.

Keywords: Stratified split, class ratio preservation

Q21. 5-fold Cross Validation kya hai? Result batao.

[IMPORTANT]

Ans: Data ko 5 equal parts mein baanto. Har baar 1 part test, baaki 4 train. 5 different results aate hain. Average lo - pata chalta hai ki model stable hai ya nahi. Hamara DT ka 5-fold CV F1 macro 0.8743 +/-

0.0155.

Example (hamare project se): Agar single split pe 0.8896 aaya aur CV pe 0.8743 - matlab model overfit nahi hai, stable hai.

Keywords: K-fold, cross validation, stability, variance

Q22. Classification aur clustering mein kya difference?

[IMPORTANT]

Ans: Classification supervised hai - labeled data, model sikh ke naye data pe label deta hai. Clustering unsupervised hai - labels nahi hote, algorithm khud se similar cheezein group karta hai.

Example (hamare project se): Humne dono kiye - pehle 3 classes pe classify kiya, phir bina label ke K-Means/DBSCAN se 5 clusters nikale.

Keywords: Supervised vs unsupervised, labeled vs unlabeled

Q23. Elbow method kya hai? k=5 kaise chuna?

[IMPORTANT]

Ans: Elbow method: different k values pe K-Means chala ke inertia (within-cluster sum of squares) plot karte hain. Jaha graph ka 'kone' (elbow) banta hai, wahi best k hota hai. Us point ke baad inertia slow hoti hai.

Example (hamare project se): Hamare elbow curve mein k=5 pe kink aayi - isliye k=5 chuna. Inertia 69,115.75.

Keywords: Elbow method, inertia, WCSS, k selection

Q24. K-Means se pehle scaling kyu karte hain?

[IMPORTANT]

Ans: K-Means Euclidean distance pe based hai. Agar ek feature 1000s mein aur doosra 10s mein ho, to bada feature dominate kar deta. Scaling se sab features barabar weight pe aate hain.

Example (hamare project se): Bina scaling, distance wala feature cluster decide kar raha hota, duration_h ka role nahi rehta.

Keywords: Euclidean distance, scaling, K-Means

Q25. Silhouette score kya batata hai?

[IMPORTANT]

Ans: Ye -1 se +1 ke beech hota hai. Batata hai ki point apne cluster mein kitna fit hai vs doosre cluster se kitna alag. +1 perfect, 0 boundary, -1 galat cluster.

Example (hamare project se): Hamara K-Means silhouette 0.2024 (fix ke baad). DBSCAN 0.2543 - thoda behtar. Low-medium hai kyunki clusters overlap karte hain real data mein.

Keywords: Silhouette, cohesion, separation, -1 to +1

Q26. K-Means vs DBSCAN - kab kaunsa?

[IMPORTANT]

Ans: K-Means: k pehle dena padta, spherical clusters, outlier-sensitive. DBSCAN: k nahi batana, density-based, arbitrary shape, noise alag kar deta. DBSCAN better jab clusters irregular ho aur outliers ho.

Example (hamare project se): K-Means (k=5) silhouette 0.2024, DBSCAN (eps=7, min_samples=20) 4 clusters + 0.4% noise, silhouette 0.2543 - DBSCAN thoda behtar.

Keywords: K-Means, DBSCAN, density-based, noise, eps, min_samples

Q27. PCA kya hai? Plot ke liye kyu use kiya?

[MEDIUM]

Ans: PCA (Principal Component Analysis) dimensionality reduction technique hai. 26 features ko 2 principal components mein compress karta hai jo maximum variance capture karte hain. Isse 2D plot bana sakte hain.

Example (hamare project se): Hum 26 features ko direct plot nahi kar sakte. PCA ne 2D mein laaya taaki clusters visually dikh sakein - clusters_pca.png mein dikhaya.

Keywords: PCA, dimensionality reduction, variance, principal components

Q28. 5 clusters ko kaise interpret kiya?

[IMPORTANT]

Ans: Har cluster ka mean nikaal ke dekha. 5 clusters: (1) ER-regional, (2) NFR-mixed, (3) Short-distance Pass, (4) Long-distance AC Premium, (5) Unknown-zone Short Pass. Zone, distance, duration, coach types se interpret kiya.

Example (hamare project se): Cluster 3 (Short Pass) mein distance kam, sleeper coaches zyada. Cluster 4 (Long AC) mein distance zyada, 3AC/2AC zyada.

Keywords: Cluster profiling, mean features, interpretation

Q29. Data quality fix ki poori story sunao (viva ke liye ready rakho).

[MOST IMP]

Ans: EDA karte waqt notice kiya ki duration_m column ka range sirf 0-58 tha, jabki duration_h alag tha. Matlab duration_m total minutes nahi, sirf minute-component tha. Fix kiya: total_duration = duration_h*60 + duration_m. Phir saare models dobara train kiye - sab ka F1 sudhar gaya. Ye project ka central finding hai: data quality > algorithm choice.

Example (hamare project se): Pehle DT 0.8309, baad 0.8896. Naive Bayes bhi 0.2969 -> 0.3098. Sab consistent improvement.

Keywords: Data quality fix, duration_m, total_duration, before/after

Q30. Isse kya seekha? Biggest takeaway?

[MOST IMP]

Ans: Biggest takeaway: Data quality > Algorithm choice. Log samajhte hain naya algorithm laane se accuracy badhegi. Lekin humne dekha ek simple fix ne 5 models ka performance badha diya. Viva mein zaroor bolna.

Example (hamare project se): Agar hum data fix na karte, DT 0.83 pe atka rehta. Kitne bhi models try karte, data hi galat tha.

Keywords: Data quality > Algorithm choice, takeaway

Q31. Har model ka fix se pehle/baad F1 batao.

[MOST IMP]

Ans: KNN 0.7552 -> 0.7886 (+0.0334). DT 0.8309 -> 0.8896 (+0.0587). DT(depth=10) 0.7720 -> 0.8459 (+0.0739). Naive Bayes 0.2969 -> 0.3098 (+0.0129). SVM 0.7168 -> 0.7651 (+0.0483). Silhouette 0.1755 -> 0.2024 (+0.0269). Sab up!

Example (hamare project se): Ye table before_after_duration_fix.png mein bhi saved hai.

Keywords: Before/after table, consistent improvement

Q32. Project ki limitations kya hain?

[IMPORTANT]

Ans: (1) Data 2020-21 ka snapshot hai, real-time nahi. (2) Sirf 3 classes cover kiye - real IR mein aur types hain. (3) SF aur Exp ka overlap hai - confusion matrix mein dikha. (4) Zone info mein kuch 'Unknown' values thi. (5) Model deployment not tested - notebook-level hai.

Example (hamare project se): SF ka error 17.4% - sabse zyada, kyunki SF aur Exp ke features overlap karte hain.

Keywords: Snapshot data, 3 classes only, SF-Exp overlap, deployment

Q33. Future mein kya improve kar sakte ho?

[MEDIUM]

Ans: (1) Real-time data integrate - API se. (2) Deep learning try - Neural Network. (3) SMOTE se minority class (SF) balance. (4) Hyperparameter tuning - GridSearchCV. (5) Streamlit dashboard. (6) Zyada classes add - Rajdhani, Shatabdi, Vande Bharat.

Example (hamare project se): SMOTE se SF ke 719 samples ko 1,288 tak la sakte hain.

Keywords: SMOTE, GridSearchCV, Streamlit, Neural Network, real-time

Q34. Tool kaunsa use kiya?

[EASY]

Ans: VS Code with Python 3.13.6 aur Jupyter extension. Notebook .ipynb format mein. Terminal PowerShell.

Example (hamare project se): Libraries: pandas, numpy, scikit-learn, matplotlib, seaborn, joblib. Sab pip se install - requirements.txt mein hai.

Keywords: VS Code, Python 3.13.6, Jupyter extension, .ipynb

Q35. Anaconda kyu nahi?

[EASY]

Ans: Anaconda ek bundled distribution hai - 3-4 GB ka. Humne Python aur libraries manually pip se install kiye. Same result, lighter setup, cleaner environment.

Example (hamare project se): pip install -r requirements.txt - bas.

Keywords: pip vs conda, lightweight setup

Q36. Ye project reproducible hai? Kaise?

[IMPORTANT]

Ans: Bilkul. (1) requirements.txt hai - pip install -r se saare libraries. (2) Data CSV repo mein hai. (3) Notebook top-to-bottom chalao. (4) Random seed fixed - same results repeat honge. (5) Models bhi saved - load karke use kar sakte ho.

Example (hamare project se): Kisi bhi laptop pe git clone, pip install, notebook run - same results.

Keywords: Reproducibility, requirements.txt, random seed, saved models

Q37. Confusion matrix ko 3 classes ke liye kaise padhte hain?

[MEDIUM]

Ans: 3x3 grid hoti hai. Row = actual class, column = predicted. Diagonal pe correct predictions (Pass-Pass, Exp-Exp, SF-SF). Off-diagonal pe confusions - humne dekha SF 25 baar Exp bola gaya, Exp 16 baar SF.

Example (hamare project se): SF ka error 17.4% highest kyunki SF aur Exp features kaafi similar hain.

Keywords: 3x3 confusion matrix, diagonal, off-diagonal

Q38. Micro vs macro vs weighted averaging mein difference?

[MEDIUM]

Ans: Macro = har class ka score ka simple average (minority ko equal weight). Weighted = class size se weight (majority dominate). Micro = total TP/FP/FN ka aggregate. Imbalance wale data mein macro best.

Example (hamare project se): Hamare DT: macro 0.8896, weighted 0.9146, micro 0.9150. Macro lowest hai kyunki SF minority hai - wahi sach batata hai.

Keywords: Macro, micro, weighted, averaging strategies

Q39. Kya model overfit hai? Kaise pata chala?

[HARD]

Ans: Nahi overfit nahi hai. Kaise pata chala: train accuracy ~ test accuracy. Aur 5-fold CV F1 0.8743 +/- 0.0155 - low variance. Agar overfit hota, train 99% test 70% hota aur CV mein bahut fluctuation aata.

Example (hamare project se): DT default parameters pe 0.9150 test accuracy diya. Depth 10 pe kam aaya (0.8859) - matlab simple model better hai.

Keywords: Overfitting, train vs test, variance, CV

Q40. Agar koi feature missing ho jaye production mein toh?

[HARD]

Ans: Options: (1) Imputation - mean/median se bharo. (2) Model-based impute - KNN imputer. (3) Us feature ko drop karke retrain. (4) Agar missing rate bahut zyada - feature hatao. Production mein imputation pipeline preprocessing ke saath attach karni chahiye.

Example (hamare project se): Hamare data mein nulls nahi the, lekin real production mein ye common problem hai - pipeline mein imputer zaroor rakho.

Keywords: Imputation, KNN imputer, preprocessing pipeline, production